Mzameli Mashiyi, AIN580S, Tut 2, BEng Tech (Computer Engineering), 2026.

Import Libraries

In [1]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

1. Read a local file called “wfa_girls_0-to-13-weeks_zscores.xlsx" into a dataframe.

In [2]:
```
# Load the Excel file into a DataFrame
df = pd.read_excel("wfa_girls_0-to-13-weeks_zscores.xlsx")
```

2. Look at the top and bottom 5 rows of the data using head() and tail().

In [3]:
```
# Look at the top and bottom 5 rows of the data using head() and tail().

# Display the first five rows
df.head(5)
```

Out[3]:

	Week	L	M	S	SD3neg	SD2neg	SD1neg	SD0	SD1	SD2	SD3
0	0	0.3809	3.2322	0.14171	2.0	2.4	2.8	3.2	3.7	4.2	4.8
1	1	0.2671	3.3388	0.14600	2.1	2.5	2.9	3.3	3.9	4.4	5.1
2	2	0.2304	3.5693	0.14339	2.3	2.7	3.1	3.6	4.1	4.7	5.4
3	3	0.2024	3.8352	0.14060	2.5	2.9	3.3	3.8	4.4	5.0	5.7
4	4	0.1789	4.0987	0.13805	2.7	3.1	3.6	4.1	4.7	5.4	6.1

In [4]:
```
# Display the last few rows
df.tail(5)
```

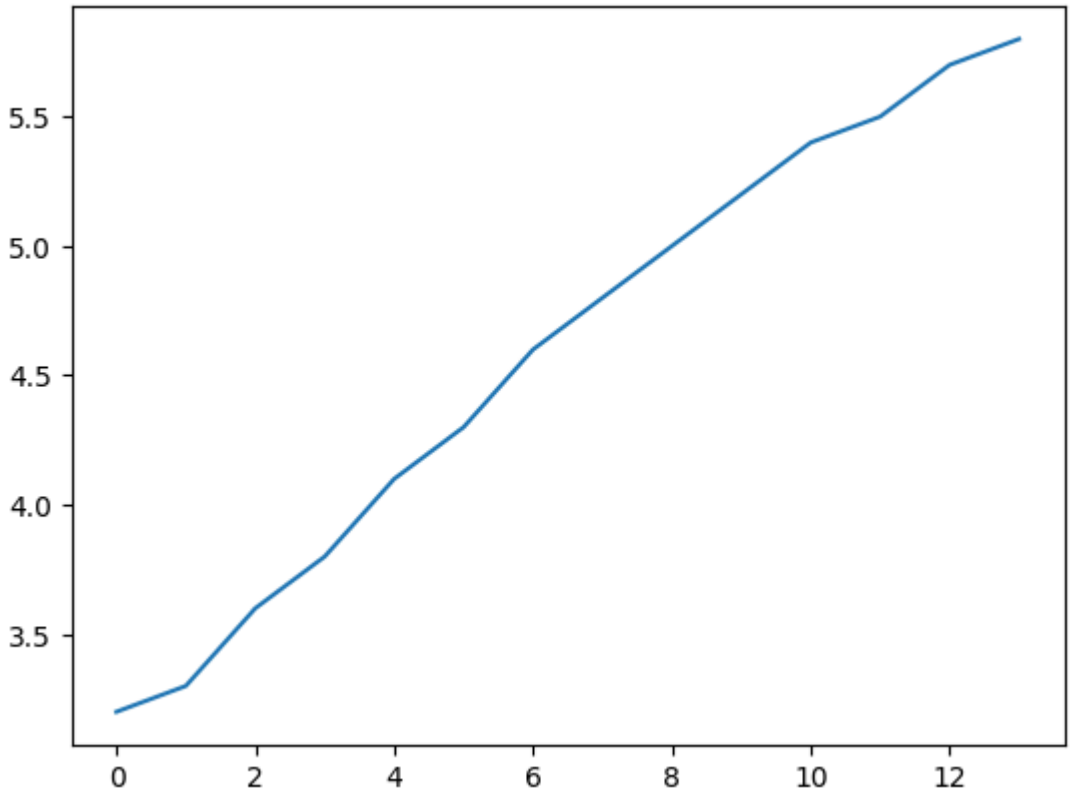
Out[4]:

	Week	L	M	S	SD3neg	SD2neg	SD1neg	SD0	SD1	SD2	SD3
9	9	0.0918	5.1842	0.12966	3.5	4.0	4.6	5.2	5.9	6.7	7.6
10	10	0.0779	5.3618	0.12861	3.6	4.1	4.7	5.4	6.1	6.9	7.8
11	11	0.0648	5.5295	0.12770	3.8	4.3	4.9	5.5	6.3	7.1	8.1
12	12	0.0525	5.6883	0.12691	3.9	4.4	5.0	5.7	6.5	7.3	8.3
13	13	0.0407	5.8393	0.12622	4.0	4.5	5.1	5.8	6.6	7.5	8.5

3. Plot Week vs SD0.

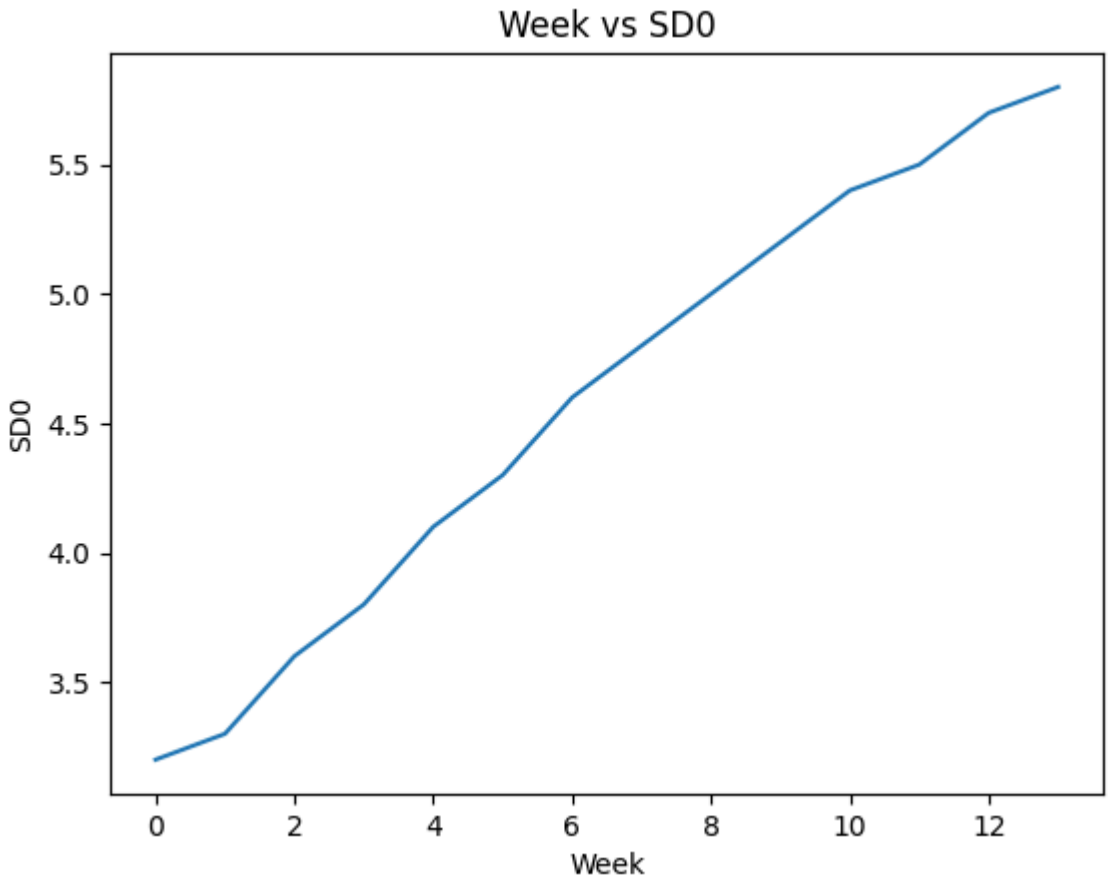
In [5]:
```
# Plot Week vs SD0
plt.figure()
plt.plot(df['Week'], df['SD0'])
```

Out[5]: [<matplotlib.lines.Line2D at 0x1e8c8d86c10>]



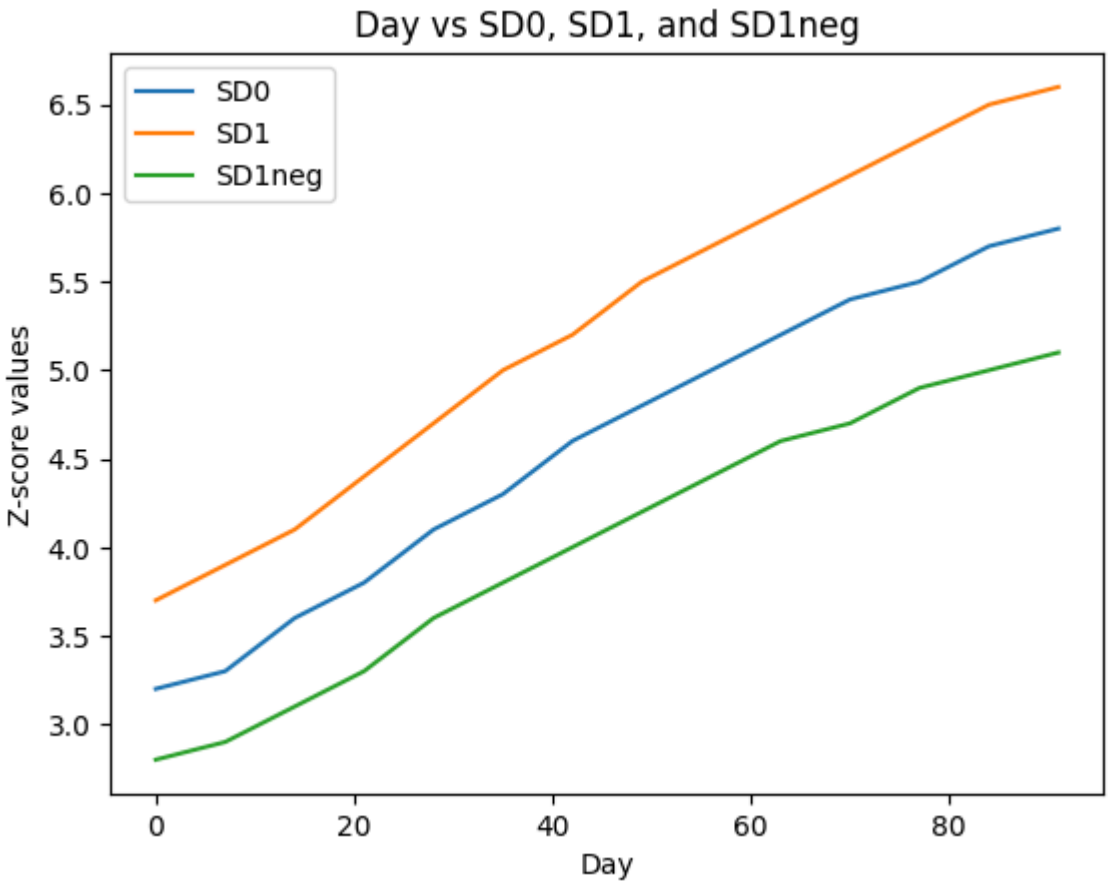
4. Assign suitable axis labels to the plot.

In [6]:
```
# Plot Week vs SD0
plt.figure()
plt.plot(df['Week'], df['SD0'])

# Add labels and title
plt.xlabel('Week')
plt.ylabel('SD0')
plt.title('Week vs SD0')

# Show the plot
plt.show()
```



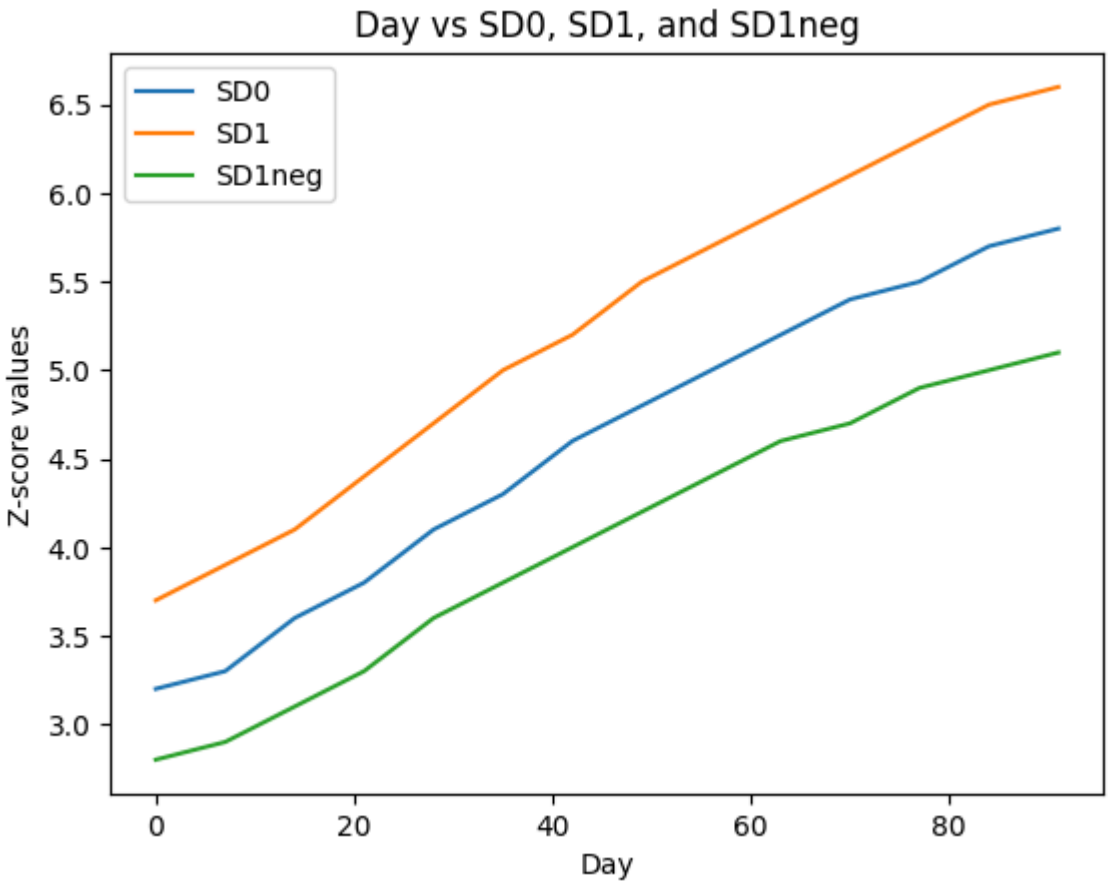
5. Plot day vs SD0, SD1, and SD1neg.

In [7]:
```
# Create a Day column from Week
df['Day'] = df['Week'] * 7

# Plot Day vs SD0, SD1, and SD1neg
plt.figure()

plt.plot(df['Day'], df['SD0'], label='SD0')
plt.plot(df['Day'], df['SD1'], label='SD1')
plt.plot(df['Day'], df['SD1neg'], label='SD1neg')

# Labels and title
plt.xlabel('Day')
plt.ylabel('Z-score values')
plt.title('Day vs SD0, SD1, and SD1neg')

plt.legend()
plt.show()
```



6. Save the plot as an image. The format of the image is unimportant, experiment with the options.

In [8]:
```
# Create a Day column from Week
df['Day'] = df['Week'] * 7

# Plot Day vs SD0, SD1, and SD1neg
plt.figure()

plt.plot(df['Day'], df['SD0'], label='SD0')
plt.plot(df['Day'], df['SD1'], label='SD1')
plt.plot(df['Day'], df['SD1neg'], label='SD1neg')

# Labels and title
plt.xlabel('Day')
plt.ylabel('Z-score values')
plt.title('Day vs SD0, SD1, and SD1neg')

plt.legend()


# Save the figure
plt.savefig('Z-score values over time.png', dpi=300, bbox_inches='tight')

plt.show()
```



7. Scale any SD column by a factor of 100.

In [9]:
```
# I tried to scale SD0, SD1, SD2

# List of SD columns
sd_columns = ['SD0', 'SD1', 'SD2']

# Scale them by 100
df[sd_columns] = df[sd_columns] * 100

# Verify the changes
print(df[sd_columns].head())
```

```
     SD0    SD1    SD2
0  320.0  370.0  420.0
1  330.0  390.0  440.0
2  360.0  410.0  470.0
3  380.0  440.0  500.0
4  410.0  470.0  540.0
```

8. Change all the column headings to lower case.

In [10]:
```
# Convert column names to lowercase
df.columns = df.columns.str.lower()

# Verify the changes
print(df.columns)
```

```
Index(['week', 'l', 'm', 's', 'sd3neg', 'sd2neg', 'sd1neg', 'sd0', 'sd1',
       'sd2', 'sd3', 'day'],
      dtype='object')
```

9. Sort all data by descending week.

In [11]:
```
# Sort by week (descending)
df = df.sort_values(by='week', ascending=False)

# View the result
print(df.head())
```

```
    week       l       m        s  sd3neg  sd2neg  sd1neg    sd0    sd1  \
13    13  0.0407  5.8393  0.12622     4.0     4.5     5.1  580.0  660.0
12    12  0.0525  5.6883  0.12691     3.9     4.4     5.0  570.0  650.0
11    11  0.0648  5.5295  0.12770     3.8     4.3     4.9  550.0  630.0
10    10  0.0779  5.3618  0.12861     3.6     4.1     4.7  540.0  610.0
9      9  0.0918  5.1842  0.12966     3.5     4.0     4.6  520.0  590.0

      sd2  sd3  day
13  750.0  8.5   91
12  730.0  8.3   84
11  710.0  8.1   77
10  690.0  7.8   70
9   670.0  7.6   63
```

10. Find the index and value of the min, mean, and max values in the S column. (See section 5.3 and 5.3 in “ch5” notebook)

In [12]:
```
# Minimum value and its index
min_value = df['s'].min()
min_index = df['s'].idxmin()

# Maximum value and its index
max_value = df['s'].max()
max_index = df['s'].idxmax()

# Mean value
mean_value = df['s'].mean()

print("Minimum value:", min_value, "at index:", min_index)
print("Mean value:", mean_value)
print("Maximum value:", max_value, "at index:", max_index)
```

```
Minimum value: 0.12622 at index: 13
Mean value: 0.13441071428571424
Maximum value: 0.146 at index: 1
```

11. Calculate the percent change of any SD column and save the result as a new dataframe column named "PCSDx".

In [13]:
```
# Use column is sd0 to calculate the percent change and create a new column call
df['PCSDx'] = df['sd0'].pct_change() * 100  # multiply by 100 to get percent
# View the new column
print(df[['sd0', 'PCSDx']].head())
```

```
      sd0     PCSDx
13  580.0       NaN
12  570.0 -1.724138
11  550.0 -3.508772
10  540.0 -1.818182
9   520.0 -3.703704
```

12. Write the dataframe to a file named “tutorial2.csv”.

In [14]:
```
df.to_csv("tutorial2.csv", index=False)
```

13. Confirm the image and csv files you generated are as you expect.

In [15]:
```
# Load the saved CSV
df_check = pd.read_csv("tutorial2.csv")

# Preview the data
print(df_check.head())

# Check structure
print(df_check.info())
```

```
   week       l       m        s  sd3neg  sd2neg  sd1neg    sd0    sd1    sd2  \
0    13  0.0407  5.8393  0.12622     4.0     4.5     5.1  580.0  660.0  750.0
1    12  0.0525  5.6883  0.12691     3.9     4.4     5.0  570.0  650.0  730.0
2    11  0.0648  5.5295  0.12770     3.8     4.3     4.9  550.0  630.0  710.0
3    10  0.0779  5.3618  0.12861     3.6     4.1     4.7  540.0  610.0  690.0
4     9  0.0918  5.1842  0.12966     3.5     4.0     4.6  520.0  590.0  670.0

   sd3  day     PCSDx
0  8.5   91       NaN
1  8.3   84 -1.724138
2  8.1   77 -3.508772
3  7.8   70 -1.818182
4  7.6   63 -3.703704
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14 entries, 0 to 13
Data columns (total 13 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   week    14 non-null     int64
 1   l       14 non-null     float64
 2   m       14 non-null     float64
 3   s       14 non-null     float64
 4   sd3neg  14 non-null     float64
 5   sd2neg  14 non-null     float64
 6   sd1neg  14 non-null     float64
 7   sd0     14 non-null     float64
 8   sd1     14 non-null     float64
 9   sd2     14 non-null     float64
 10  sd3     14 non-null     float64
 11  day     14 non-null     int64
 12  PCSDx   13 non-null     float64
dtypes: float64(11), int64(2)
memory usage: 1.6 KB
None
```

14. Export your notebook to pdf. If you are using VSCode. Go to extensions. Install print. Look at the top of your code. Look for export. You should be able to export the code to PDF. If there are issues with converting to PDF, convert it to HTML then to PDF.